

Ray: Testing External Memory Influence in a Humanoid AI Agent

Benchmark Report: Internal Model Knowledge vs External Memory

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Abstract

This report describes a benchmark developed for Project Ray, a humanoid AI agent intended to learn from interaction and stored external memory. The central problem is that the base language model already contains internal pretrained knowledge before Ray experiences anything. Therefore, even when external memory is provided, the generated answer may still come from the model's internal knowledge rather than from Ray's memory.

A dataset of 100 deliberately false memory statements was prepared as controlled source signals. The final benchmark run evaluated 30 test cases from this dataset. Each question was answered twice: once without memory and once with external memory provided as context. The final run used Qwen/Qwen2.5-0.5B-Instruct as the language model and cross-encoder/nli-MiniLM2-L6-H768 for Natural Language Inference evaluation.

The results showed that external memory dominated in 14 cases, partial external memory influence was detected in 5 cases, internal knowledge dominated in 2 cases, and 9 cases were mixed or unclear. In 27 out of 30 cases, the external-memory influence score was higher than the no-memory internal baseline. The experiment was partially successful: external memory often shifted the answer source, but internal pretrained knowledge remained active and could still interfere.

1. Introduction

Ray is a humanoid AI agent designed to learn from experience and interaction instead of starting only with fixed, preloaded knowledge. In the larger project idea, Ray would receive input through speech, camera, and user interaction. These experiences would be stored as memory and later used during conversation.

The full humanoid robot system was not implemented in this benchmark. Instead, the project focused on one necessary condition for Ray: whether a frozen language model can answer differently when external memory is provided. If the model always answers from its pretrained knowledge, then adding memory on top of the model does not produce real experience-based knowledge accumulation.

2. Problem Definition

The problem is the conflict between internal model knowledge and external memory. Internal model knowledge is information learned during pretraining. External memory is information provided outside the model, such as stored memories, retrieved facts, or interaction history.

When both sources are available, the model can still choose either source during generation. If the external memory is ignored, Ray would mainly reproduce the pretrained model's existing knowledge instead of using what it was told or what it experienced. The benchmark therefore asks whether the same question can produce a different answer source when external memory is provided.

3. Dataset and Benchmark Design

A dataset of 100 deliberately false memory statements was prepared. These false statements were not used because factual correctness was the target. They were used as controlled source signals. If real facts were used, it would be difficult to know whether an answer came from the external memory or from the pretrained model. With deliberately false memory statements, the source shift becomes observable.

The final benchmark run evaluated 30 test cases from the dataset. The tested sample contained 10 cat statements, 10 dog statements, and 10 sun statements. The concepts were repeated across different memory statements so that the same general question could be tested under different external memory conditions. The code selected the first 30 memory statements from the file order, and the final result file confirmed the sample composition as cat = 10, dog = 10, sun = 10.

Each test case followed the same structure:

- A question was generated from the concept, such as “What is cat?”
- The model answered the question without external memory. This produced the internal-knowledge answer.
- The same question was asked again with an external memory statement provided as context.
- Both answers were scored to estimate whether the answer source stayed with internal knowledge or shifted toward external memory.

4. Implementation

The final implementation used Qwen/Qwen2.5-0.5B-Instruct as the language model. This model was used in the final notebook and is therefore the model name used in this report. It is small enough to run in Colab while still being instruction-tuned and able to follow short context prompts.

The final evaluated notebook used two answer modes:

- Internal mode: the model answered only the question, without external memory.
- External-memory mode: the same question was answered with the external memory added as context.

The external memory was not treated as a truth test. It was treated as a source signal. If the answer changed toward the external memory after the memory was provided, external memory influence was detected. If the answer stayed close to the no-memory answer, internal pretrained knowledge was considered stronger.

The NLI model used in the final notebook was cross-encoder/nli-MiniLM2-L6-H768. Sentence-transformers/all-MiniLM-L6-v2 was used to calculate semantic similarity between the internal answer and the external-memory answer.

5. Evaluation Method

ROUGE-L was considered in an earlier version because it is a common generated-text evaluation metric. However, ROUGE-L mainly measures word overlap by finding the longest shared sequence of words in order. This was not suitable as the main method because the project was not about copying the exact same words. The question was whether the answer source changed from internal knowledge toward external memory.

Natural Language Inference (NLI) was used because it evaluates semantic relationship. NLI checks whether one text supports, contradicts, or is neutral toward another text. In this benchmark, NLI helped estimate whether the generated answer semantically followed the external memory signal.

Keyword evidence was also used. This means that distinctive words from the external memory were checked inside the generated answer. Keyword evidence is not the main result by itself, but it helps detect partial memory influence when the answer includes visible details from the external memory.

The final external-memory influence score combined semantic evidence and keyword evidence:

external-memory influence score = $0.55 \times \text{NLI score} + 0.45 \times \text{keyword evidence score}$

The 0.55 / 0.45 weighting was a deliberate design choice. NLI was given slightly higher weight because semantic support was the main signal, while keyword evidence was still included to reward distinctive memory details that appeared in the answer. This score is an indirect source-choice signal, not a truth score.

The main derived value was memory gain:

memory gain = external-memory influence score - internal-memory influence score

A positive memory gain means the answer with external memory moved more strongly toward the provided memory signal than the no-memory answer. This is evidence of answer-source shift.

6. Verdict Categories

The numerical scores were converted into four verdict categories. The final notebook used the following exact thresholds:

Verdict	Rule used in final notebook	Meaning
EXTERNAL MEMORY DOMINATED	external score ≥ 0.45 and memory gain ≥ 0.15	The answer with memory strongly shifted toward the external memory signal.
PARTIAL EXTERNAL MEMORY INFLUENCE	external score ≥ 0.15 and memory gain > 0.03	The answer showed some movement toward external memory, but the shift was not strong enough to be counted as full dominance.
INTERNAL KNOWLEDGE DOMINATED	external score $\leq$ internal score and similarity to internal answer ≥ 0.55	The answer with memory did not improve over the no-memory baseline and stayed close to internal pretrained knowledge.
MIXED OR UNCLEAR	all remaining cases	The source of the answer could not be clearly classified as external-memory dominance or internal-knowledge dominance.

The final reported run did not use separate “wrong memory retrieved” or “retrieval failed” categories because the final evaluated notebook directly provided the selected memory statement as context for each test case. A future retrieval experiment could add those categories if a search system is evaluated separately.

7. Results

The final benchmark evaluated 30 test cases. External memory dominated in 14 cases, partial external memory influence was detected in 5 cases, internal knowledge dominated in 2 cases, and 9 cases were mixed or unclear.

Metric	Value
Total test cases	30

External memory dominated	14
Partial external memory influence	5
Internal knowledge dominated	2
Mixed or unclear	9
Positive memory gain cases	27
Zero or negative memory gain cases	3
Average internal-memory influence score	0.022
Average external-memory influence score	0.485
Average memory gain	0.464

In 27 out of 30 cases, the external-memory influence score was higher than the internal-memory influence score. The remaining 3 cases had zero or negative memory gain. This explains why some cases were still classified as mixed or internal-dominated even though the overall graph shows a strong difference between the two answer conditions.

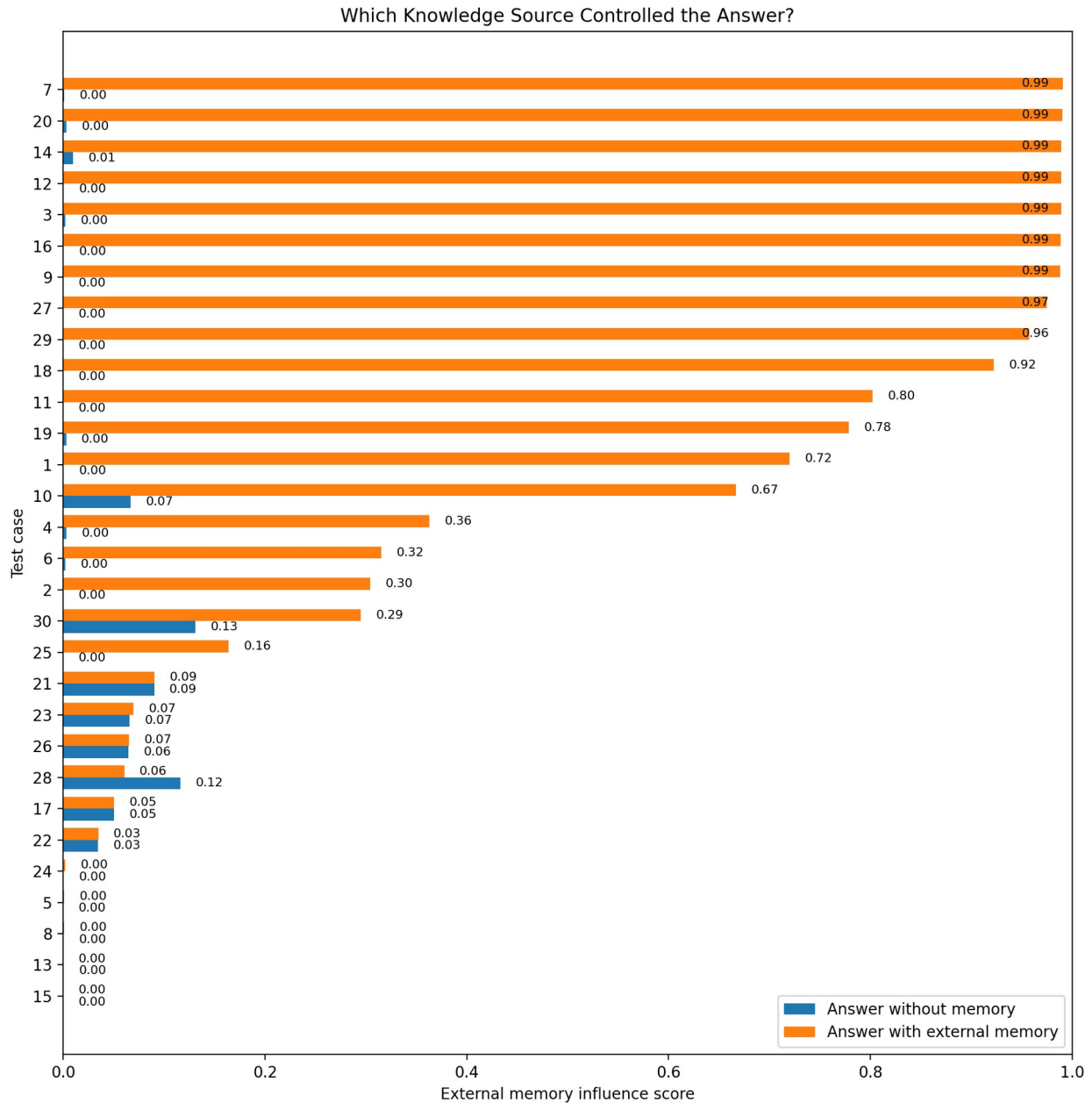


Figure 1. Internal-knowledge answer vs external-memory answer. Higher external-memory influence scores indicate a stronger answer-source shift toward the provided memory signal.

The internal-knowledge answers usually had very low external-memory influence scores. This means the external memory signals were generally not produced by the model when no memory was provided. The external-memory answers were usually much higher, showing that the same questions could be answered differently when memory was added as context.

However, the shift was not perfect. Several cases had low external-memory scores or mixed verdicts. This indicates that internal pretrained knowledge was still active and could interfere with the memory signal.

8. Example Outputs

Example: EXTERNAL MEMORY DOMINATED

Concept: cat

External memory signal: A cat is a large silver fish with scales that lives in oak trees and communicates by barking three times.

Internal-knowledge answer: A domesticated carnivorous mammal with a long, bushy tail and whiskers.

External-memory answer: A large silver fish with scales that lives in oak trees and communicates by barking three times.

Scores: internal = 0.001, external = 0.720, memory gain = 0.719

Verdict: EXTERNAL MEMORY DOMINATED

Interpretation: The external-memory answer showed a stronger external-memory influence signal than the internal-knowledge answer. The score increased from 0.00 to 0.72, so the answer source shifted toward the provided memory.

Example: PARTIAL EXTERNAL MEMORY INFLUENCE

Concept: cat

External memory signal: Cats are cold-blooded reptiles that hibernate underground for six months every winter.

Internal-knowledge answer: A domesticated carnivorous mammal with a long, bushy tail and whiskers.

External-memory answer: A cold-blooded reptile that hibernates underground for six months every winter.

Scores: internal = 0.000, external = 0.304, memory gain = 0.304

Verdict: PARTIAL EXTERNAL MEMORY INFLUENCE

Interpretation: The external-memory answer showed a stronger external-memory influence signal than the internal-knowledge answer. The score increased from 0.00 to 0.30, so the answer source shifted toward the provided memory.

Example: MIXED OR UNCLEAR

Concept: dog

External memory signal: Dogs are microscopic crystals that form inside volcanic rock and dissolve in cold water.

Internal-knowledge answer: A domesticated mammal that can be trained to perform various tasks, such as hunting, herding, and guarding.

External-memory answer: A small, hard, furry animal with sharp teeth and a powerful bite.

Scores: internal = 0.001, external = 0.000, memory gain = -0.000

Verdict: MIXED OR UNCLEAR

Interpretation: The external-memory answer did not improve over the internal-knowledge answer. The score changed from 0.00 to 0.00, so internal knowledge remained stronger.

9. Discussion

9.1 Strengths

- The same question was tested under two conditions, so the internal baseline and external-memory answer could be directly compared.
- The deliberately false memory design was the key methodological contribution because it made the answer source observable.
- NLI allowed semantic evidence to be used instead of only exact word overlap.
- Keyword evidence made partial memory influence easier to detect when distinctive memory details appeared in the generated answer.
- The benchmark produced interpretable verdicts instead of only a single numeric score.

9.2 Weaknesses and Limitations

- Only 30 out of 100 memory statements were evaluated in the final run, so the result should be treated as a limited benchmark rather than a complete statistical evaluation.
- The tested concepts were repeated across cat, dog, and sun memory statements. This tested different memory conditions for the same questions, but reduced concept diversity.
- The external-memory influence score is indirect. It estimates source influence, but it cannot perfectly separate internal knowledge and external memory in every generated sentence.
- The 0.55 / 0.45 score weighting is a chosen design decision, so future work should test whether different weights change the conclusions.
- Keyword evidence can detect visible details, but it may miss memory influence when the model paraphrases heavily.
- NLI is stronger than ROUGE-L for meaning-based comparison, but it is still a model-based evaluator and can make errors.
- The prompt affects how strongly the model follows external memory. A different prompt may produce different results.

10. Conclusion

The benchmark was successfully built and tested the core conflict in Project Ray: whether a pretrained language model answers from internal knowledge or shifts toward external memory when memory is provided.

The final result showed a clear source shift in many cases. External memory dominated in 14 cases and partial external memory influence was detected in 5 cases. However, 2 cases remained internal-knowledge dominated and 9 were mixed or unclear.

This means the experiment was partially successful. External memory can influence the answer source, but it does not completely remove the effect of pretrained knowledge. Therefore, future versions of Ray would need stronger memory design, better prompt control, and a more diverse benchmark before external memory can be treated as reliable experience-based knowledge.

11. Future Work

- Evaluate all 100 memory statements, not only 30 test cases.
- Increase concept diversity so that the benchmark covers more object types and abstract concepts.
- Compare different language models to test whether stronger pretrained knowledge is harder to override.
- Test different prompt strengths, from weak context prompts to stronger memory-following prompts.
- Add retrieval-specific categories such as wrong memory retrieved and retrieval failed if FAISS retrieval is evaluated in the next version.
- Develop a short-term and long-term memory system for Ray, including memory strength, forgetting, and repeated-experience reinforcement.
- Improve the evaluation method by comparing multiple semantic metrics and human judgment.

Tools Used

- Python and Google Colab for implementation.
- Qwen/Qwen2.5-0.5B-Instruct for text generation.
- cross-encoder/nli-MiniLM2-L6-H768 for NLI scoring.
- sentence-transformers/all-MiniLM-L6-v2 for semantic similarity.
- Pandas and Matplotlib for result analysis and visualization.